



HAIR DISORDERS

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By the end of the lecture the student will be able to:

1-Mention the main hair disorders.

2-Mention the definition and clinical picture of androgenic alopecia.

3- Mention the definition and clinical picture of telogen effluvium.

4-Mention the definition and clinical picture of alopecia areata.

5-Identify the etiology and the main methods of treatment of the main hair disorders.

Source: AMBOSS Medical Learning Platform.

CONTENTS:

1-Androgenetic Alopecia.

2-Alopecia Areata.

3-Telogen Effluvium .

1-ANDROGENETIC ALOPECIA

Definition:

A progressive, non-scarring alopecia that affects the regions of the scalp with the most androgen-sensitive hair follicles, resulting in a characteristic pattern of balding (bitemporal scalp in men and vertex and frontal scalp in women).



Epidemiology

- **Most common type of alopecia**
- **Prevalence: Increases with age**
- **Sex: ♂ > ♀ (but very common in women as well)**
- **Age at onset: puberty in males; menopause in females**

  .

- **Ethnicity: more common in white populations**

Etiology

- **Increased androgen sensitivity of hair follicle in genetically predisposed individuals .**

- **Uncommon causes:**
  - **Congenital adrenal hyperplasia**
  - **Androgen-producing tumors: Leydig cell tumor.**

Differential diagnosis

- **Diffuse Alopecia Areata**

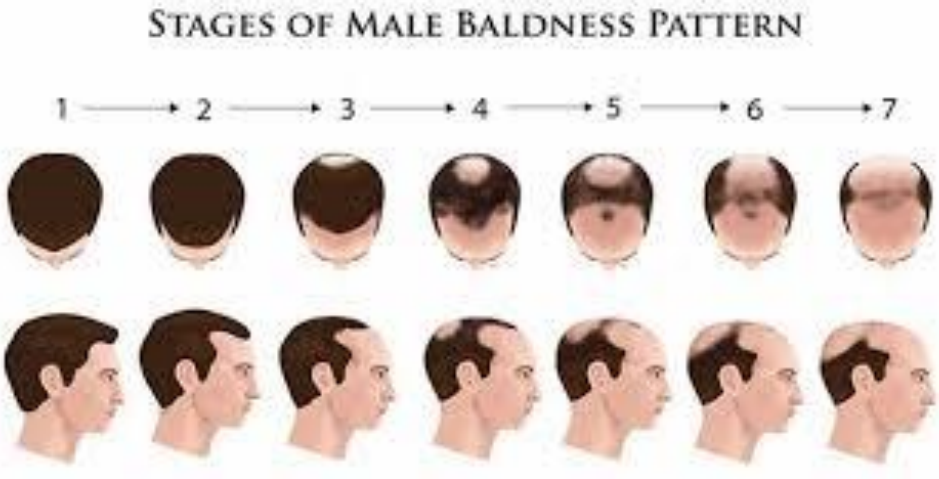
- **Telogen effluvium**

Androgenetic alopecia can be present in both men **(Male pattern baldness)** and women **(Female pattern baldness)** with minimal difference between them in clinical features and treatment.

1-Male Pattern Baldness:

Clinical features:

- **Gradual, nonscarring hair loss**
- **Bitemporal "M" pattern of recession**
- **May be followed by hair loss on the vertex of the scalp**











Treatment:

- **First-line treatments:**

Finasteride

5% minoxidil (direct arteriolar vasodilator; can also be used for refractory hypertension) .

- **Hair transplant surgery:**

Follicular units from the occipital scalp are extracted (either as small units or as a linear strip), divided into small units, and implanted into the bald areas.

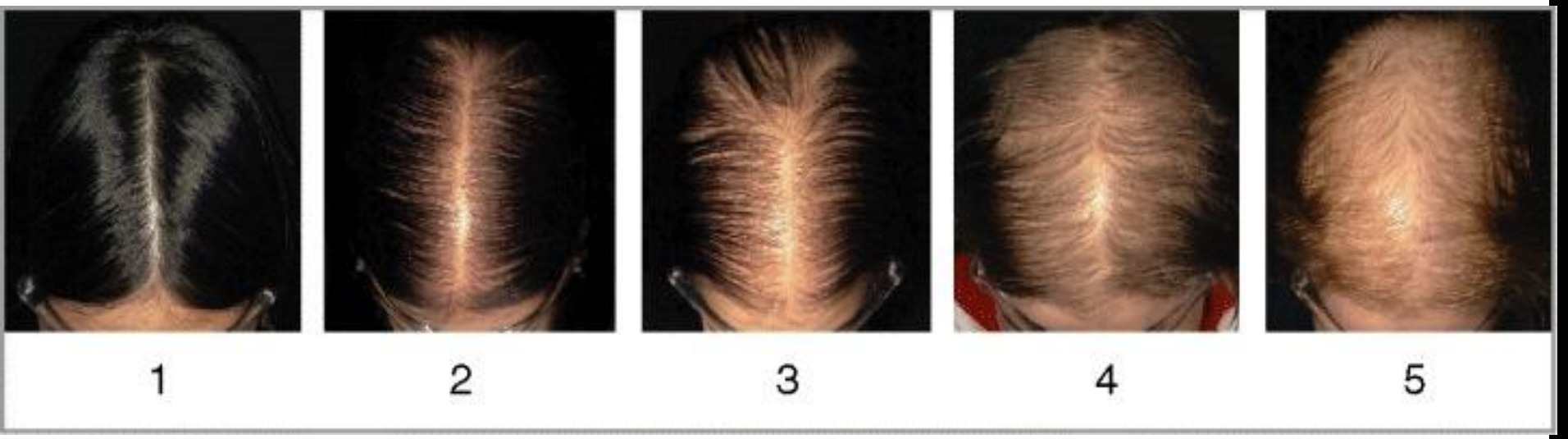
- **Camouflage:**

Keratin fibers, hair dyes, .. etc.

2-Female Pattern Baldness

Clinical features:

- **Gradual, non-scarring hair loss.**
- **Begins at the vertex.**
- **Progresses to a more diffuse hair thinning of the entire scalp.**
- **Seen mainly in post-menopausal women.**
- **Hirsutism may also be present.**
- **Seen also in women with hyperandrogenism (due to PCOS, androgen secreting ovarian tumors, etc.): acne, hirsutism, menstrual irregularities, infertility, etc.**


1
2
3
4
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Treatment:

- **First-line Treatment:**

Topical Minoxidil 2%

- **Oral Anti-androgens:**
  - Spironolactone: First choice for systemic therapy
  - Finasteride
  - Flutamide
  - Cyproterone acetate:
- **Surgery and camouflage.**

2-ALOPECIA AREATA

Definition: **Hair loss in well demarcated patches due to immune-mediated inflammation of hair follicles.**



Epidemiology

- Prevalence: 1 in 1000 people
- Age: mostly in people < 30 years
- Sex: ♂ = ♀



Etiology:

- Immune mediated inflammation and disruption of anagen phase hair follicles → well-defined patches of nonscarring hair loss
- A trigger factor (emotional stress, infections, pregnancy, etc.) may precede some cases
- Family history in up to 20% of cases

Clinical features :

Abrupt onset (within weeks).

Smooth, circular, well defined patches of hair loss without scarring .

Various patterns of distribution :

- **Extensive alopecia areata:** hair loss affecting > 50% of the scalp
- **Alopecia universalis :** All hair-bearing sites are affected.
- **Alopecia totalis:** Complete baldness .
- **Nail involvement** (up to 40% of cases): nail pitting, onycholysis, etc.
- Other autoimmune disorders may be present (e.g., vitiligo, autoimmune disorders of the thryoid, etc.)

Alopecia Totalis



Alopecia Universalis



Differential diagnosis

Trichotillomania: Compulsive pulling out of one's own hair → ill-defined patchy hair loss and hairs of different lengths.

- Tinea capitis: The affected areas of scalp are scaly compared to the smooth hair loss of alopecia.
- Secondary syphilis: patchy hair loss → moth-eaten appearance of scalp.

Treatment

- Intralesional steroids (triamcinolone) .
- Topical immunotherapy (DPCP (diphenylcyclopropenone) or SADBE (squaric acid dibutyl ester)
- PUVA treatment (Psoralen + UVA)

Prognosis:

Poor for alopecia universalis and totalis

3-TELOGEN EFFLUVIUM

- **Telogen effluvium is the loss of hair in response to an overwhelming physiologic stress, such as cancer or malnutrition.**
- **Treatment is correction of the underlying stress or disease.**

Etiology

1-Endocrine: hypothyroidism, hypopituitarism, hypoparathyroidism, hyperthyroidism, diabetes mellitus.

2- Drugs: anabolic steroids, antithyroid drugs, heparin, antimetabolites (chemotherapy), oral contraceptives, etc.

3-Nutritional: hypervitaminosis A, zinc deficiency, malnutrition states.

4-Stress: major surgery, sepsis, after childbirth (telogen gravidarum), acute or chronic illness (e.g., systemic lupus erythematosus), severe psychological stress.

Mechanism:

The telogen (resting) phase of hair growth predominates → premature shedding of the resting hair follicles → diffuse, nonscarring hair loss.

Clinical features :

- **Hair density reduces by < 50%**
- **Does not progress to complete baldness.**

Treatment:

Reassurance and manage underlying condition


THANK YOU
FOR YOUR
TIME